Pearson Correlation for Ordinal Variables: Consolidated Analysis

Table of contents

1 Extremal Pearson Correlation for Ordinal Variables

1.1 1. Problem Setting and Notation

We consider two ordinal variables:

- X with K_X ordered categories $0, 1, \dots, K_X - 1$
- Y with K_Y ordered categories $0, 1, \dots, K_Y - 1$

Marginals given by:

$$p_X(i) = P(X = i), \quad p_Y(j) = P(Y = j)$$

or counts:

$$n_{X=i}, \quad n_{Y=j}, \quad N = \sum_i n_{X=i} = \sum_j n_{Y=j}$$

CDFs:

$$F_X(i) = \sum_{k \le i} p_X(k), \quad F_Y(j) = \sum_{k \le j} p_Y(k)$$

Pearson correlation:

$$r_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad E[XY] = \sum_{i=0}^{K_X - 1} \sum_{j=0}^{K_Y - 1} x_i y_j \, P(X = x_i, Y = y_j)$$

For Spearman's correlation, replace X, Y by their ranks R_X, R_Y.

(Sources: Proofs-OrdinalVariables.md; Ordinal Variables and Correlation.md; BFbounds.md; Math Proof GPT – Whitt 1976 Paper.md)

1.2 2. Fréchet–Hoeffding / Boole–Fréchet Bounds

Any joint CDF F_{XY} with marginals F_X, F_Y satisfies:

$$\max\{F_X(i) + F_Y(j) - 1, 0\} \;\le\; F_{XY}(i, j) \;\le\; \min\{F_X(i), F_Y(j)\}$$

(BFbounds.md)

- **Upper bound (comonotonic)**:

$$F_{XY}^{\text{upper}}(i, j) = \min\{F_X(i), F_Y(j)\}$$

- **Lower bound (countermonotonic)**:

$$F_{XY}^{\text{lower}}(i, j) = \max\{F_X(i) + F_Y(j) - 1, 0\}$$

1.3 3. Maximizing and Minimizing r

1.3.0.1 3.1 Max r — Comonotonic Coupling

Claim: The maximum correlation $r_{\max}$ is achieved by the comonotonic (FH upper) arrangement.

Proofs / Arguments: - **FH upper pmf construction**: In discrete case, differences of CDF steps allocate mass along the diagonal, pairing high X with high Y.
(BFbounds.md; Proofs-OrdinalVariables.md) - **Rearrangement inequality (ranks)**: For ranks R_X, R_Y, $\sum R_X R_Y$ is maximized when sequences are ordered the same way.
(Ordinal Variables and Correlation.md) - **Quantile coupling**: $U \sim \text{Unif}(0,1)$, $X = F_X^{-1}(U), Y = F_Y^{-1}(U)$.
(Math Proof GPT – Whitt 1976 Paper.md)

1.3.0.2 3.2 Min r — Countermonotonic Coupling

Claim: The minimal correlation $r_{\min}$ corresponds conceptually to the countermonotonic (FH lower) arrangement.

Proofs / Arguments: - **FH lower pmf construction**: Differences of the lower-bound CDF.
(BFbounds.md) - **Opposite-order pairing**: Sort X descending, Y ascending; achieves best feasible negative correlation.
(Proofs-OrdinalVariables.md; Ordinal Variables and Correlation.md) - **Quantile coupling**: $X = F_X^{-1}(U), Y = F_Y^{-1}(1-U)$ in ideal continuous case.
(Math Proof GPT – Whitt 1976 Paper.md)

Note: In discrete settings, FH lower may not be attainable; see Section 5.

1.4 4. Binary (2×2) Cases and Examples

(From Cheemera v1.1 – Pearson Correlation Analysis Implications.md)

1.4.0.1 4.1 Balanced marginals (70–30, 70–30)

- Perfect diagonal fill yields $r_{\max} = 1$.

1.4.0.2 4.2 Skew vs balanced (90–10, 50–50)

- Perfect alignment impossible; $r_{\max} < 1$ due to capacity limits.

1.5 5. Attainability and Symmetry (Updated)

1.5.0.1 5.1 Maximal correlation

- **Result:** In the discrete ordinal setting, $r_{\max}$ is **always** attained by the FH upper (comonotonic) construction.
  (BFbounds.md; Proofs-OrdinalVariables.md; Ordinal Variables and Correlation.md)

1.5.0.2 5.2 Minimal correlation

- **Result:** The FH lower (countermonotonic) construction may be **unattainable** in discrete settings; $r_{\min}$ is then the best feasible opposite-order arrangement.
  (BFbounds.md; Proofs-OrdinalVariables.md; Ordinal Variables and Correlation.md)

1.5.0.3 5.3 Magnitude relationships

- **Earlier view:** Often $|r_{\min}| < r_{\max}$ for discrete asymmetrical marginals.
  (BFbounds.md; Ordinal Variables and Correlation.md)
- **New evidence (BES data)**: Possible to have $|r_{\min}| > r_{\max}|$.
  Example: X with 4 categories, Y with 5 categories, skewed marginals;
  $r_{\max} = 0.85027$, $r_{\min} = -0.89916$.
  (Cees – r_min MAGNITUDE Asymmetry in Pearson Correlation.md)

Reason: Pearson correlation uses centered values; skewness can make negative deviations in antitone pairing larger in magnitude than positive deviations in comonotone pairing.

1.5.0.4 5.4 Symmetry condition

- If one marginal is **palindromically symmetric** about its center rank, then $r_{\min} = -r_{\max}$.
  (Cees – r_min MAGNITUDE Asymmetry in Pearson Correlation.md)

1.6 6. Simulation / Randomization

Permutation distribution:
Generate full vectors from marginals, independently permute X and Y, compute r, repeat. Produces null distribution centered at 0.
(Proofs-OrdinalVariables.md)

1.7 7. Key Tools Referenced

- **Fréchet–Hoeffding bounds** for extremal joint CDFs.
  (BFbounds.md)
- **Hardy–Littlewood–Pólya rearrangement inequality** for rank-based maximization/minimization.
  (Ordinal Variables and Correlation.md)
- **Comonotone/countermonotone quantile coupling**.
  (Math Proof GPT – Whitt 1976 Paper.md)
- **Capacity limits** in discrete joint tables explain why FH lower may be unattainable.

1.8 8. Master Takeaways

1. $r_{\max}$ **= FH upper (comonotonic), always attainable** in discrete ordinal settings.
2. $r_{\min}$ may **not** equal FH lower in discrete settings; use best feasible opposite-order arrangement.
3. No universal bound on relative magnitudes: $|r_{\min}|$ may be less than, equal to, or greater than $r_{\max}$.
4. Palindromic symmetry of one marginal guarantees $|r_{\min}| = r_{\max}|$.
5. Binary cases illustrate when perfect alignment is feasible and when skew prevents it.

---

2 Positive Semi-Definiteness in Covariance Matrices: Implications for PCA, Factor Analysis, and SEM

2.1 1. Variance–Covariance Matrices: Definition and Core Property

2.1.0.1 1.1 Definition

For a random vector $\mathbf{X} = (X_1, X_2, \dots, X_n)^\top$, the variance–covariance matrix is

$$= \mathbb{E}\left[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top\right],$$

where $\sigma_{ij} = \mathrm{Cov}(X_i, X_j)$.
(vcov_PSD.md):contentReferenceoaicite:2

- **Diagonal elements** σ_{ii} are variances, always ≥ 0.
- **Off-diagonal** entries are covariances.

2.1.0.2 1.2 PSD Property

A matrix is **positive semi-definite (PSD)** if, for all nonzero $\mathbf{z}$,

$$\mathbf{z}^\top \ \mathbf{z} \geq 0.$$

For covariance matrices:

$$\mathbf{z}^\top \ \mathbf{z} = \mathbb{E}\big[(\mathbf{z}^\top(\mathbf{X} - \mathbb{E}[\mathbf{X}]))^2\big] \geq 0,$$

since it is an expectation of a square.
(vcov_PSD.md):contentReferenceoaicite:3

2.1.0.3 1.3 Positive Definite (PD) vs PSD

- **PD** if $\mathbf{z}^\top \ \mathbf{z} > 0$ for all $\mathbf{z} \neq 0$, i.e., no perfect linear dependence among variables.
- **PSD** if at least one nonzero $\mathbf{z}$ yields equality (rank-deficient case). *(vcov_PSD.md):contentReferenceoaicite*

2.2 2. PCA and PSD Requirements

2.2.0.1 2.1 PCA Overview

Principal Components Analysis (PCA) decomposes a sample covariance or correlation matrix $\mathbf{S} = n^{-1}\mathbf{X}^\top\mathbf{X}$ into

$$\mathbf{S} = \mathbf{Q}\,\mathrm{diag}(\lambda_1, \dots, \lambda_p)\,\mathbf{Q}^\top.$$

(PCA and FA PSD requirement.md):contentReferenceoaicite:5

2.2.0.2 2.2 Why PSD Matters for PCA

- **Mathematically**: Eigen-decomposition works for any symmetric $\mathbf{S}$, PSD or not.
- **Interpretively**: Negative eigenvalues break the "variance explained" story (variance cannot be negative).
- **Software behavior**: Most PCA routines check PD/PSD; failure can cause warnings or dropped components. *(PCA and FA PSD requirement.md):contentReferenceoaicite:6*

2.2.0.3 2.3 If Input is Symmetric but not PSD

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad \lambda = (3, -1),$$

is indefinite (one negative eigenvalue).
- PCA math still runs; eigenvalues/vectors computed. - Interpretation fails: $\sqrt{\lambda_j}$ imaginary for $\lambda_j < 0$. - Some software refuses input; others run but return nonsensical variance metrics. *(PCA and FA PSD requirement.md):contentReferenceoaicite:7*

2.2.0.4 2.4 Remedies for PCA

1. Treat tiny negative eigenvalues as zero.

2. Adjust to nearest PSD matrix (Higham 2002).

3. Run SVD on the raw data matrix instead of precomputed $\mathbf{S}$.
   (PCA and FA PSD requirement.md):contentReferenceoaicite:8

2.3 3. Factor Analysis and PSD Requirements

2.3.0.1 3.1 FA Model

Common factor analysis models Σ as:

$$\Sigma = \Lambda\Lambda^\top + \Psi,$$

where Ψ is diagonal with positive entries. *(PCA and FA PSD requirement.md):contentReferenceoaicite:9*

2.3.0.2 3.2 PSD/PD Needs in FA

- **ML FA**: Requires $\mathbf{S} \succ 0$; $\log\det\mathbf{S}$ undefined if indefinite.
- **Principal-axis / Image FA**: Also break with negative eigenvalues (square roots fail).
  (PCA and FA PSD requirement.md):contentReferenceoaicite:10

2.3.0.3 3.3 Remedies for FA

- Repair $\mathbf{S}$ to PSD before analysis.
- Add small ridge term $\epsilon\mathbf{I}$. *(PCA and FA PSD requirement.md):contentReferenceoaicite:11*

2.4 4. SEM and PSD Requirements

2.4.0.1 4.1 SEM Machinery

Observed covariance **S**; implied covariance $\Sigma(\theta)$ from structural and measurement models. *(PCA and FA PSD requirement.md):contentReferenceoaicite:12*

2.4.0.2 4.2 Fit Functions and PD Needs

- **ML / GLS**: Require both **S** and $\Sigma(\theta)$ PD.
- **WLS/DWLS**: Need PD indirectly (weight matrix invertible).
- **ULS**: Can start from indefinite **S**, but test statistics again require PD. *(PCA and FA PSD requirement.md):contentReferenceoaicite:13*

2.4.0.3 4.3 What Happens if Not PSD

- Software errors out ("matrix is not positive definite").
- Log-determinant undefined; inversion fails.
- Even if bypassed, optimisation steps hit complex/NaN values. *(PCA and FA PSD requirement.md):contentReferenceoaicite:14*

2.4.0.4 4.4 Remedies in SEM

- Nearest PD smoothing (Higham 2002).
- Ridge adjustment.
- Analyse raw data.
- Check rounding issues. *(PCA and FA PSD requirement.md):contentReferenceoaicite:15*

2.5 5. Core Lessons Across PCA, FA, SEM

1. **Theoretical PSD**: Covariance matrices from real data are PSD by definition:contentReferenceoaicite:16.
2. **Numerical Issues**: Sampling noise, rounding, or listwise deletion can yield small negative eigenvalues:contentReferenceoaicite:17.
3. **PCA**: Can compute from indefinite matrix but interpretation breaks:contentReferenceoaicite:18.
4. **FA and SEM**: Require PD input; will not run on indefinite matrices:contentReferenceoaicite:19.
5. **Repairs**: Adjust to nearest PSD, ridge, or recompute from raw data:contentReferenceoaicite:20.
